UNIT III

Safety Acts, Industrial Hygiene & Occupational Health

Factories Act | Explosives Act | Boilers Act | ESI Act | Workmen's Compensation | hygiene | ergonomics | diseases | noise | codes

I. Features of the Factories Act, 1948

Objective: to **regulate working conditions** in factories - **health, safety, welfare, hours of work, leave and employment of young persons**. Extends to the whole of India; applies to factories with 10+ workers (power) or 20+ (no power). Enforced by State Governments.

Feature	Provision
Health (Ch. III)	Cleanliness (Sec 11), waste disposal (12), ventilation & temperature (13), dust & fume (14), humidification (15), overcrowding (16 - 14.2 m3/worker), lighting (17), drinking water (18), latrines & urinals (19), spittoons (20).
Safety (Ch. IV)	Fencing of machinery (21), work on or near machinery in motion (22), young persons on dangerous machines (23), striking gear & belts (24), cranes & lifts (28, 29), revolving machinery (30), pressure plant (31), floors, stairs & access (32), pits & sumps (33), excessive weights (34), eye protection (35), dangerous fumes (36), explosive dust (37), fire precaution (38), safety of buildings (40).
Hazardous processes (IVA)	Site appraisal committee (41A), compulsory disclosure of information (41B), specific responsibility of occupier (41C), medical examination, permissible limits (41F), workers' participation (41G), right to warn (41H).
Welfare (Ch. V)	Washing facilities (42), storing clothing (43), sitting (44), first aid (45), canteen (46), shelters (47), creche (48), welfare officer (49).
Working hours (Ch. VI)	48 hrs/week, 9 hrs/day, spread-over 10.5 hrs, interval 30 min after 5 hrs, weekly holiday, double wages for overtime, no double employment.
Young persons (Ch. VII)	No child below 14; adolescents 15-18 need fitness certificate and token; only day shift 6 am-7 pm (children 4.5 hrs).
Women	Employment in night shift (7 pm-6 am) allowed only with consent and safeguards; no work with dangerous machines / hazardous substances in some cases.

Feature	Provision
Leave (Ch. VIII)	Annual leave with wages: 1 day per 20 days worked (adult), 1 per 15 (child).
Penalty (Ch. X)	Fine and / or imprisonment; enhanced penalty for repeated offences and where death or injury results.

Note

> Since 21 Nov 2025 these provisions are carried in the **OSH Code, 2020**. For the syllabus, Factories Act, 1948 is the reference.

II. The Explosives Act, 1884

An Act to regulate the **manufacture, possession, use, sale, transport and import** of explosives (gunpowder, dynamite, detonators, fireworks, nitrates...) in India. The present rules are **Explosives Rules, 2008**, administered by **PESO** (Petroleum & Explosives Safety Organisation), Nagpur, under DPIIT.

> **Licence** is required to manufacture, possess, sell, transport or use explosives; granted by the Chief Controller / District Magistrate.

> Explosives are classified (Class 1 divisions and safety distances): **1.1 mass explosion, 1.2 projection, 1.3 fire, 1.4 no significant hazard**.

> Safe distances between buildings (inhabited, magazine, process) are fixed in the rules; storage in **magazines** with earth-mound, lightning protection and no iron tools.

> Rules on **packing, labelling, transport** (vehicle marking, speed, escort), prohibition of smoking, fire, matches near explosives.

> Power to prohibit especially dangerous explosives; duty to **report** accidents; inspectors can enter, inspect, seize.

> Penalty: imprisonment up to 3-5 years and/or fine; offences under the Act are cognisable in some cases.

Related laws

> **Explosive Substances Act, 1908** - criminal law against misuse of explosives.

> **Petroleum Act, 1934** and **Gas Cylinder Rules** - petroleum & compressed gas storage/transport.

III. The Indian Boilers Act, 1923

Provides for the **safety of life and property** from **explosions of steam boilers** by laying down registration, inspection and certification. Rules: **Indian Boiler Regulations (IBR), 1950**. Boiler = closed vessel of capacity above **22.75 litres** that generates steam under pressure above atmospheric pressure from heat of fuel or other source.

Aspect	Provision
Authority	Central Boilers Board (makes regulations), **Chief Inspector of Boilers** in each State, Inspectors of Boilers.
Registration	A boiler cannot be used unless registered with a number & **certificate** issued after inspection; maximum working pressure is fixed.
Inspection	Annual (internal & external) inspection and hydraulic test; after repair or change; certificate valid for 12 months.
Mountings	Safety valves (2), water level gauges (2), pressure gauge, stop valve, feed check valve, blow-down valve, fusible plug, manhole.
Operation	Boiler attendants / **boiler operators** must hold a certificate of competency; not to be operated beyond rated pressure.
Accidents	Every explosion or serious damage must be reported to the Inspector; no repair of the wreckage till inspected.
Materials & repairs	Boiler materials, welding of pressure parts, welders and repairs must meet IBR; approved manufacturers.
Penalty	Use of unregistered or uncertified boiler is punishable by fine and imprisonment; explosion due to negligence - severe.

IV. Employees' State Insurance (ESI) Act, 1948

A social-security Act that gives **medical care and cash benefits** to employees in case of **sickness, maternity, employment injury, disablement and death**. Run by the **ESI Corporation** (ESIC); funded by contributions.

Point	Detail
Coverage	Factories and notified establishments with **10 or more** employees; employees earning up to **Rs 21,000** per month (Rs 25,000 for disabled).

Point	Detail
Contribution	Employer **3.25%** + employee **0.75%** of wages (total 4%); employees earning up to Rs 176/day exempt from their share.
Sickness benefit	Cash about **70% of wages** for up to 91 days in a year, plus full medical care.
Maternity benefit	Full wages for 26 weeks (first two children) and 12 weeks thereafter.
Disablement benefit	Temporary: 90% of wage during disablement. Permanent: pension based on loss of earning capacity.
Dependants' benefit	Monthly pension (about 90% of wage) to widow, children if the worker dies of employment injury.
Other	Medical benefit (for self & family), funeral expenses (Rs 15,000), unemployment allowance (Rajiv Gandhi Shramik Kalyan Yojana).

> Employee covered by ESI is **not** entitled to compensation under the Workmen's Compensation Act for the same injury.

> Administration: **ESIC** (corporation), standing committee, medical benefit council; ESI hospitals and dispensaries.

> Note: after 21 Nov 2025 the ESI Act is subsumed in the **Code on Social Security, 2020**; benefits remain broadly the same.

V. Workmen's Compensation Act, 1923

Now called the **Employees' Compensation Act (2009 amendment)**. Employer is liable to pay **compensation** if a worker suffers **personal injury by accident arising out of and in the course of employment** (resulting in death or disablement) or an **occupational disease** listed in Schedule III.

Point	Provision
Liability	Employer pays even without fault (no-fault liability) - employer not liable if the injury is due to drunkenness, wilful disobedience of safety rules or removal of safety device.
Death	50% of monthly wages x relevant factor (age-wise); minimum Rs 1,20,000.
Permanent total disablement	60% of monthly wages x relevant factor; minimum Rs 1,40,000.
Partial / temporary	Percentage of the above according to loss of earning capacity (Schedule I); temporary disablement: half-monthly payment.

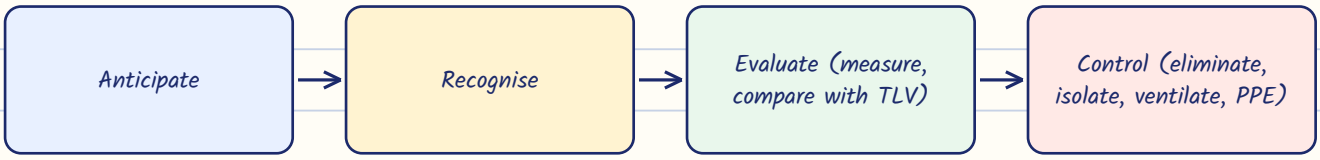
Point	Provision
Wage ceiling	For calculation, monthly wage taken is capped at Rs 8,000 (as revised by Govt).
Procedure	Claim to the **Commissioner for Workmen's Compensation** within 2 years; employer deposits compensation; penalty and interest for delay.
Medical	Employer bears medical expenses; notice of accident to be given to employer.

Compare - ESI Act vs Workmen's Compensation Act

Point	ESI Act	WC Act
Type	Social insurance (contributory)	Employer's liability (no premium)
Benefit	Medical + cash for sickness, maternity, injury	Cash on injury / disease / death
Covers	Employees up to wage ceiling	All workmen (with limits)
Administered by	ESIC	State Commissioner

VI. Industrial hygiene

Industrial hygiene is the science and art of **anticipation, recognition, evaluation and control** of environmental factors at work that may cause sickness, impaired health, discomfort or inefficiency.



Hazard class	Examples
Chemical	Dust, fumes, mists, gases, vapours - silica, lead, benzene, chlorine, CO
Physical	Noise, vibration, heat/cold, radiation, illumination, pressure
Biological	Bacteria, virus, fungi - anthrax, tetanus, TB in hides, hospitals
Ergonomic	Poor posture, repetitive motion, heavy lifting, bad workstation
Psychosocial	Stress, shift work, harassment, long hours

Threshold Limit Values (TLV, ACGIH)

> **TLV-TWA**: time-weighted average concentration for an 8-hr day / 40-hr week to which nearly all workers can be exposed repeatedly without harm.

> **TLV-STEL**: 15-min short-term exposure limit - not more than 4 times a day, at least 60 min apart.

> **TLV-C (Ceiling)**: concentration that should never be exceeded, even instantaneously.

> Units: **ppm** for gases/vapours, **mg/m3** for dust/fumes, f/cc for fibres. Factories Act 2nd Schedule lists permissible levels.

$$TWA = (C1\ T1 + C2\ T2 + \ldots + Cn\ Tn) / 8$$

Eg: 2 hrs at 100 ppm, 6 hrs at 20 ppm -> TWA = (200 + 120)/8 = **40 ppm**.

VII. Occupational safety, health and disease prevention

Occupational health (WHO/ILO): promotion and maintenance of the highest degree of **physical, mental and social well-being** of workers in all occupations.

> **Pre-employment & periodic medical examination**; special tests (lung function, audiometry, blood lead).

> **Identify and control** hazards at source; limit exposure time; job rotation.

> **Education**, personal hygiene (wash hands, no eating at work), vaccinations.

> Health records, **health surveillance**, rehabilitation and suitable job after illness.

> Occupational health centre, doctor, nurse; ambulance and tie-up with hospital.

> Levels of prevention: **primary** (remove cause), **secondary** (early detection), **tertiary** (treatment & rehabilitation).

VIII. Occupational diseases

A disease that arises mainly from work exposure - listed in **Schedule III** of the Factories Act / WC Act (e.g. 29 listed diseases).

Disease	Cause / industry	Effect & prevention
Silicosis	Free silica dust - mining, quarry, foundry, sand-blasting	Fibrosis of lungs; wet drilling, LEV, respirators, medical test
Asbestosis	Asbestos fibres - insulation, brake lining	Lung scarring, cancer; ban/replace, enclose
Byssinosis	Cotton dust - textile mills	'Monday fever', breathlessness; dust control
Coal worker's pneumoconiosis	Coal dust - coal mines	Black lung; dust suppression

Disease	Cause / industry	Effect & prevention
Lead poisoning	Lead - battery, paint, printing	Anaemia, nerve damage; substitution, hygiene, blood test
Noise-induced hearing loss	Noise > 85 dB - weaving, forging	Permanent deafness; control noise, ear plugs, audiometry
Dermatitis	Oils, acids, cement, chromium	Skin rash; gloves, barrier creams, washing
Anthrax	Hides, wool, bones - tanneries	Skin/lung infection; vaccination, disinfection
Hand-arm vibration (white finger)	Pneumatic tools, chainsaws	Numb blanching fingers; anti-vibration tools, limit time
Cadmium / benzene / mercury	Plating, chemical industry	Kidney, blood, nervous system; closed system, monitoring

IX. Ergonomics

Ergonomics (human factors) = fitting the **job to the worker** instead of making the worker fit the job - it joins anatomy, physiology and psychology with engineering design to give comfort, safety, health and efficiency.

Area	Principle / example
Workstation design	Adjustable seat, work surface at elbow height, tools within normal reach (inner 40 cm arc), foot rest.
Posture	Neutral spine, wrists straight; avoid static, twisted or overhead postures; change posture often.
Manual handling	Keep load close to body, limit weight (about 25 kg), team lift, use trolley/hoist.
Repetitive work	Rotate tasks, rest pauses, power tools with low vibration - avoid RSI, carpal tunnel syndrome.
Controls & displays	Standard colour (red stop, green start), logical layout, readable dials, proper labels, error-proofing.
Environment	Right light (200-500 lux), temperature 20-26 C, low noise and vibration.

> Benefits: fewer **musculoskeletal disorders** (MSDs), less fatigue, fewer errors and accidents, higher productivity.

> Anthropometry: design for 5th to 95th percentile of the user population.

X. Stress and fatigue

	Stress	Fatigue
Meaning	Body's response to demands/pressures that exceed coping ability	Reduced capacity to work due to prolonged mental or physical effort
Causes	Overload, deadlines, role conflict, danger, poor relations, job insecurity, shift work	Long hours, heavy work, monotony, poor sleep, heat, noise, poor nutrition
Effects	Anxiety, hypertension, ulcer, errors, absenteeism, substance use	Slow reaction, errors, accidents, sickness, low output
Control	Job design, clear roles, counselling, yoga, communication	Rest pauses, shift design, rotation, canteen food, medical check

> Physical fatigue is cured by rest; **mental fatigue** needs a change of activity / sleep; chronic fatigue needs medical attention.

XI. Health, safety and the physical environment

Factor	Effect and recommended control
Illumination	Poor light causes eye strain and accidents. General 150-300 lux; fine work 750-1000 lux (IS 3646). Avoid glare; clean luminaires; emergency lights.
Ventilation	Remove heat, fumes; natural + mechanical; air change rate 6-12/hr (workshops); local exhaust for contaminants; 14.2 m3 space/worker.
Temperature & humidity	Comfort zone 20-26 C, 40-60% RH. Heat stress: wet bulb globe temperature (WBGT) index; remedies - air showers, shielding, water & salt, rest, acclimatisation. Cold: clothing, heaters.
Radiation	Ionising (X-ray, gamma): time, distance, shielding; dosimeters; AERB limit 20 mSv/yr. Non-ionising (UV, IR, laser): screens, goggles.
Vibration	Whole-body and hand-arm; isolators, damping, job rotation.
Sanitation	Drinking water, latrines, washing, waste disposal (Sec 18, 19).

XII. Engineering methods of controlling chemical hazards

Control hazards in this order of preference - **at the source** first, then along the path, last at the worker.

Method	Description / example
Substitution	Replace hazardous material by a safer one (benzene -> toluene, asbestos -> glass wool, solvent-based -> water-based paint).
Process change	Change the process to avoid the hazard (brush painting instead of spraying, wet grinding instead of dry).
Isolation / enclosure	Enclose the process, glove boxes, remote control, distance, barriers.
Local exhaust ventilation (LEV)	Hood + duct + fan + air cleaner captures contaminant at the source; capture velocity 0.5-1 m/s.
General (dilution) ventilation	Fresh air dilutes low-toxicity contaminants; not for highly toxic or high-emission.
Wet methods	Water spray to suppress dust (wet drilling in mines).
Good housekeeping & maintenance	Prevent leaks, clean spills, vacuum (not sweep), closed containers.
Administrative	Limit exposure time, job rotation, training, labels, monitoring, medical checks.
PPE (last)	Respirators, gloves, suits when other controls are not enough.

XIII. Control of industrial noise and protection against it

Noise is unwanted sound. Measured in **decibels dB(A)** on a sound-level meter; sound level is logarithmic.

Exposure for long periods causes **noise-induced hearing loss**, stress, communication problems and accidents.

Sound level dB(A)	Permitted exposure per day (Indian rules / OSHA type, 5 dB exchange)
90	8 hrs
95	4 hrs
100	2 hrs
105	1 hr
110	30 min
115	15 min

> Action level 85 dB(A) (hearing conservation programme); Factories Act permissible limit 90 dB(A) for 8 hrs; never more than 115 dB(A).

> Typical levels: conversation 60 dB, lorry 85 dB, power press 100 dB, jet take-off 120+ dB (pain 130).

Control level	Methods
At source	Silent machine, maintenance, lubrication, balancing, vibration isolators, damping, mufflers on exhaust, replace metal-to-metal impact (rubber, plastic).
Along path	Enclosures, acoustic barriers, absorbent lining, sound-proof control rooms, increase distance, layout planning.
At receiver	Ear plugs (NRR 20-30 dB), muffs, job rotation, limit exposure time, regular **audiometry**, training.

Combining two sound levels: L = 10 log10 (10^(L1/10) + 10^(L2/10)); two 90 dB machines = 93 dB

XIV. Codes and regulations for worker safety, health and systems

Code / standard	Scope
IS 14489	Code of practice on occupational safety and health audit.
IS 14490 / IS 18001	OHS management systems (now aligned with ISO 45001).
IS 5216	Guide for safety procedures in hazardous chemical handling.
ISO 45001:2018	OH&S management system - PDCA model, leadership, worker participation, risk-based.
OSHA 29 CFR 1910	US general-industry standards - machine guarding, LOTO, hazard communication.
NFPA codes	NFPA 70 (electric), 101 (life safety), 30 (flammable liquids).
Factories Rules / OSH Code rules	State rules for safety officers, medical examination, hazardous process, licences.

Codes for safety of systems (system safety)

> **IEC 61508 / 61511**: functional safety of electrical/electronic/programmable systems; **SIL (Safety Integrity Level) 1-4** for safety instrumented systems in process industries.

> **ASME Boiler & Pressure Vessel Code, API standards**: design & inspection of boilers, pressure vessels, tanks, piping.

> **ISO 12100**: machinery safety - risk assessment & risk reduction; **ISO 13849**: control systems of machines.

> Hazard study tools: **HAZOP, FMEA, FTA, What-if, Checklist, LOPA**; Safety case & safety management for major hazard installations.

XIV-A. Worked examples & extra notes

Eg 1 - Employees' compensation (death)

A worker aged 30 earning Rs 8,000/month (wage ceiling) dies in an accident. Assume the relevant factor for his age is 200 (illustrative - take the real value from Schedule IV). Compensation = 50% x 8,000 x 200 = **Rs 8,00,000**. (Higher than the minimum Rs 1,20,000, so this amount is payable.)

Eg 2 - ESI contribution

Employee wage Rs 18,000/month: employer share 3.25% = Rs 585; employee share 0.75% = Rs 135; total = **Rs 720** per month.

Eg 3 - Noise addition

Three identical machines each give 85 dB at a point. L = 85 + 10 log10(3) = 85 + 4.8 = **89.8 dB**. So 3 machines add only about 5 dB to one.

Eg 4 - Room ventilation (air changes)

Workshop 20 m x 10 m x 5 m = 1000 m3 needs 8 air changes/hr -> airflow = 8000 m3/hr = **2.22 m3/s**; choose a fan of this capacity (add loss factor).

Types of airborne contaminants

Form	Definition	Example
Dust	Solid particles made by crushing, grinding	Silica, coal, cotton dust
Fume	Very fine solid particles from condensation of vapour (<1 micron)	Welding fume, lead fume
Mist	Liquid droplets	Acid mist, oil mist
Fog / Smoke	Condensed liquid / incomplete combustion particles	Smoke from fire
Vapour	Gaseous form of a substance that is liquid at room temperature	Benzene, acetone
Gas	Substance that is gaseous at room temperature	CO, Cl2, H2S

Routes of entry of toxic substances

> **Inhalation** (main route, ~90%), **skin absorption**, **ingestion**, injection (rare).

> Effect depends on **dose = concentration x time**, toxicity, individual sensitivity.

> Acute (short, high dose) vs chronic (long, low dose); local (irritation) vs systemic (organ damage).

Hearing conservation programme

1. Measure noise (noise survey, dosimetry) and make noise-contour map; mark areas.
2. Control noise by engineering means first.
3. Provide ear protection and train in use where >85 dB(A).
4. **Audiometric testing**: baseline and annual tests; compare for shifts.
5. Keep records and review the programme.

XV. Quick revision & important questions

Remember

> Factory: 10 / 20 workers. Overcrowding 14.2 m3 / worker. First aid 1 box / 150. Canteen 250. Creche 30 women. Welfare officer 500.

> Boiler > 22.75 L; annual inspection; IBR 1950. Explosives Rules 2008 - PESO.

> ESI: 3.25% + 0.75%; limit Rs 21,000. WC: death 50%, PTD 60% x factor.

> Noise: 90 dB(A) for 8 hrs, 5 dB exchange. TWA, STEL, Ceiling (TLV).

> Control order: substitute -> engineer (LEV) -> administrative -> PPE.

1. Explain the salient features of the Factories Act, 1948. (10 M)
2. Write short notes on Explosives Act and Boilers Act. (5 M each)
3. Discuss benefits under the ESI Act. Compare with WC Act. (10 M)
4. What is industrial hygiene? Explain TLV and TWA with an example. (10 M)
5. Explain engineering methods of controlling chemical hazards. (10 M)
6. Describe noise control and protection measures. (10 M)
7. Write short notes: ergonomics, stress & fatigue, occupational diseases. (5 M each)